

TÜİK Happiness by Education Level Analysis

Applied Statistics Project • GitHub-Compatible Documentation and Technical Report

R Project

Statistical Modeling

Hypothesis Testing

TÜİK Data 2004-2022

✦ 1. About the Project

This project analyzes, using statistical methods, the emotional states (Happy, Neutral, Unhappy) of individuals across different education levels and the effect of education level on happiness, based on the 2004–2022 household survey data published by the Turkish Statistical Institute (TÜİK). The project was carried out by a team of five people.

Main Objectives

- Compare the happiness rates of higher-education graduates, high-school graduates, primary-school graduates, and individuals who did not complete a school.
- Evaluate changes in emotional state over time using parametric and non-parametric hypothesis tests.
- Explain the effect of education levels on the overall happiness score using a Multiple Linear Regression model.

R Packages Used

```
# Loading and activating the required libraries
library(readxl)      # Reading the Excel dataset
library(tidyverse)   # Data manipulation and visualization (ggplot2, dplyr)
library(knitr)        # Reporting tools
library(kableExtra)  # Advanced table formatting
library(flextable)   # Flexible table creation
library(car)          # Regression analysis and statistical tests
library(ggpubr)      # QQ-plot and normality plots
```

📁 2. Data Import and Preprocessing

The dataset was imported into R using the `readxl::read_excel()` function; variable types were inspected and categorical variables were put into a form suitable for analysis.

```
# Loading the data
projeson <- read_excel("projeson.xlsx")

# Inspecting the first and last 6 rows (using the pipe operator)
projeson |> head() |> flextable()
projeson |> tail() |> flextable()

# Converting the variable type to factor
projeson$`Emotional State` <- as.factor(projeson$`Emotional State`)
str(projeson)
```

📊 3. Descriptive Statistics and Visualization

Measures of central tendency and dispersion were calculated for the percentages of emotional state by education level.

```
# Mean, Median, and Standard Deviation calculations
mean(projeson$Yuksekokretim)
median(projeson$Yuksekokretim)
sd(projeson$Yuksekokretim)

# Summary statistics and quartiles
summary(projeson$Yuksekokretim)
quantile(projeson$`Did Not Complete a School`)
```

Visualization Methods

- Dot Plot:** `geom_point()` was used to compare the emotional-state distributions of Primary/Secondary School and Higher

Education graduates between 2004 and 2022.

- **Boxplot:** Emotional-state distributions by year were examined with `group_by()` and `geom_boxplot()`; it was observed that the happiness rate decreased and the unhappiness rate increased over time.
- **Bar and Line Charts:** Trends were visualized using `barplot()` and `plot(type="o")`.

🔗 4. Hypothesis Tests and Normality Check

4.1. Normality Check

The normality assumption for the Higher Education group and other groups was examined using a Density Plot, QQ Plot, and the Shapiro-Wilk test.

- **H0:** The data follow a normal distribution.
- **H1:** The data do not follow a normal distribution.

```
# Shapiro-Wilk Normality Test
Yuksekgretim_2004 <- projeson |> filter(Yil == 2004) |> select(Yuksekgretim)
shapiro.test(Yuksekgretim_2004$Yuksekgretim)
```

Test Result

The Shapiro-Wilk test yielded **p < 0.05**, and H0 was rejected. Although the parametric test condition was theoretically not satisfied, a T-Test was applied for comparison purposes.

4.2. Two-Sample Mean Comparison (T-Test)

The mean emotional-state scores of Higher Education graduates in 2004 and 2022 were tested (H0: $\mu_{2004} = \mu_{2022}$).

```
# T-Test for equal and unequal variance cases
t.test(x = projeson$Yuksekgretim[projeson$Yil == 2004],
       y = projeson$Yuksekgretim[projeson$Yil == 2022],
       alternative = "two.sided", var.equal = TRUE)
```

Result: $p > 0.05$ was found and H0 could not be rejected. At the 5% significance level, no statistically significant difference was found in the mean emotional state of Higher Education graduates between 2004 and 2022.

4.3. Population Proportion Tests

- **Single Population Proportion:** The equality of the 2022 Higher Education happiness proportion to 0.5 was tested (`prop.test`); with $p < 0.05$, the difference was found to be significant.
- **Two Population Proportions:** It was confirmed (H1: $p_1 > p_2$) that the 2022 Higher Education "neutral" emotional-state proportion (0.387) is greater than the 2004 proportion (0.256) ($p < 0.05$).

🔗 5. Chi-Square (χ^2) Distribution and Independence Tests

The existence of a relationship between education levels and emotional states was analyzed using Chi-Square tests.

Analysis Topic	Test Type	P-Value	Result ($\alpha = 0.05$)
2004 Higher Education - Primary School Emotional State	Chi-Square Goodness-of-Fit Test	< 0.05	H0 Rejected (Relationship exists)
2022 Education Level & Emotional State	Independence Test	< 0.05	H0 Rejected (Dependent)
2004 Education Level & Emotional State	Independence Test	< 0.05	H0 Rejected (Dependent)
Years (2004, 2012, 2022) & Emotional State	Independence Test (xtabs)	< 0.05	H0 Rejected (Dependent)

📊 6. F-Distribution Variance Comparison

The equality of variances of the "Happy" and "Unhappy" emotional states within the Higher Education group was tested (H0:

$\sigma^2_{\text{Happy}} = \sigma^2_{\text{Unhappy}}$.

```
# Variance Test
test <- var.test(x = Happy, y = Unhappy, ratio = 1, alternative = "two.sided")
# F = 5.6631, p-value = 0.1216
```

Assessment: Since $p = 0.1216 > 0.05$, H_0 cannot be rejected. There is no significant difference between the variances.

 7. Linear Regression Analysis

The categorical emotional-state variable was converted into a numerical score (Happy = 1.0, Neutral = 0.5, Unhappy = 0.0). All education levels were included in the model as independent variables.

```
# Building the Regression Model
model <- lm(Emotional_State_Numeric ~ Yuksekogretim + Lise_ve_dengi_okul +
            Ilkokul_veya_ortaokul + Ilkokul + Bir_Okul_Bitirmede, data = veri)
summary(model)
```

0.9874
R-SQUARED (R²)

141.5
F-STATISTIC

2.84e-08
MODEL P-VALUE

Model Coefficients and Significance Assessment

Variable	Coefficient (β)	P-Value	Status (α = 0.05)
Higher Education	-0.0157	0.0445	Statistically Significant
High School and Equivalent	+0.0031	0.7275	Not Significant
Primary or Secondary School	+0.0042	0.7178	Not Significant
Primary School	-0.0211	0.0211	Statistically Significant
Did Not Complete a School	+0.0015	0.9144	Not Significant

 8. Conclusion and Overall Evaluation

Summary Findings

- In the analyses conducted on TÜİK data, a statistically significant relationship was found between education level and individuals' emotional states.
- Chi-Square tests show that both education levels and years have a dependent effect on the distribution of emotional states.
- The linear regression model explains 98.74% of the variance among the variables, revealing that Higher Education ($p \approx 0.044$) and Primary School ($p \approx 0.021$) graduation rates in particular have a determining effect on the happiness score.